



2nd International Conference on **Quantum Engineered Sensing and Information Technology**

June 29 - July 02, 2026 | Telecom Paris, Paris, France

WELCOME MESSAGE:

Recent advancements in Quantum Science and Technology have significantly surpassed expectations, positioning quantum-based solutions at the forefront of addressing critical societal challenges in the 21st century—including those in telemedicine, secure communications, and intelligent systems. Among these technologies, Free-Space Optical (FSO) communication has emerged as a vital enabler for ultra-high-speed, multi-gigabit-per-second (Gb/s), large-capacity data transfer. By utilizing laser-based signal carriers, FSO Laser Communications (Laser-Com) deliver secure, high-bandwidth, wireless, line-of-sight connectivity between remote locations—offering a transformative alternative to traditional RF-based communication.

In the era of data-intensive applications and global connectivity, the explosive growth of internet and multimedia services is straining existing telecom infrastructures. Infrared (IR) laser transmitters are now being explored as a low-risk, high-reward upgrade path—providing unmatched bandwidth, low power consumption, small form factor, and enhanced resistance to interception compared to conventional RF systems.

Simultaneously, emerging quantum-enabled fields such as spintronics, neuromorphic engineering, and quantum materials are redefining the design of next-generation electronic and photonic devices. These technologies utilize the quantum properties of particles—such as spin, entanglement, and superposition—to go beyond classical device limitations and introduce entirely new paradigms in computation, sensing, and communication.

While nature offers us a rich palette of atomic elements, it is quantum engineering that allows us to rearrange these building blocks in novel, intelligent ways to create functional materials and devices not found in nature. Nowhere is this more evident than in semiconductor quantum structures, where atoms from the periodic table are combined with atomic precision to create materials with engineered optical, electronic, and spin properties.

Advanced material growth techniques—guided by first-principles simulations and precision atomic layering—are now enabling the realization of quantum devices engineered atom-by-atom. These include high-performance photonic structures capable of converting electrical energy into coherent light or detecting photons across an ultra-wide spectral range, from deep ultraviolet (DUV) to terahertz (THz) (0.2–300 microns). Furthermore, seamless integration of these devices with Silicon Photonics platforms opens the door to scalable, CMOS-compatible quantum technologies for real-world applications.

This historic conference will spotlight cutting-edge breakthroughs and future trends in quantum sensing, photonics, and nanoscale engineering. It will showcase how a deep understanding of quantum systems—inspired by nature but designed beyond it—is enabling humanity to develop powerful tools to not only coexist with, but harness, the forces of nature.




Yours Sincerely

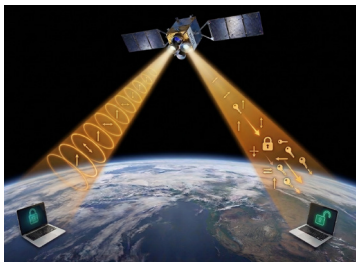
Prof. Manijeh Razeghi

Walter P. Murphy Professor and Director Center for Quantum Devices,
Department of Electrical and Computer Engineering Northwestern University, USA

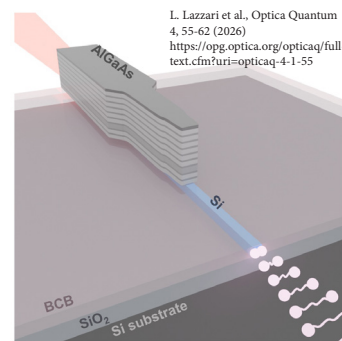
<https://www.quest-conference.com/>



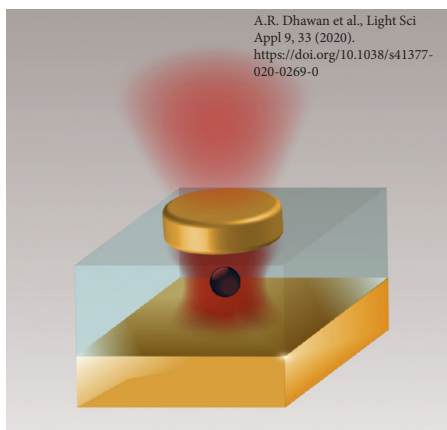
Quantum science is undergoing a profound transformation, driven by our growing ability to engineer, control, and exploit quantum systems at the most fundamental level. Across platforms ranging from integrated photonic chips and single quantum emitters to superconducting circuits and atomic-scale probes, researchers are harnessing entanglement, superposition, and quantum coherence to push the boundaries of what is physically possible. The Quantum Engineered Sensing and Information Technology (QUEST) conference brings together the communities advancing these frontiers — quantum computing, quantum communications, quantum sensing, and quantum simulation — united by the shared conviction that the deliberate engineering of quantum states will define the next generation of science and technology. Quantum science is undergoing a profound transformation, driven by our growing ability to engineer, control, and exploit quantum systems at the most fundamental level. Across platforms ranging from integrated photonic chips and single quantum emitters to superconducting circuits and atomic-scale probes, researchers are harnessing entanglement, superposition, and quantum coherence to push the boundaries of what is physically possible. The Quantum Engineered Sensing and Information Technology (QUEST) conference brings together the communities advancing these frontiers — quantum computing, quantum communications, quantum sensing, and quantum simulation — united by the shared conviction that the deliberate engineering of quantum states will define the next generation of science and technology.



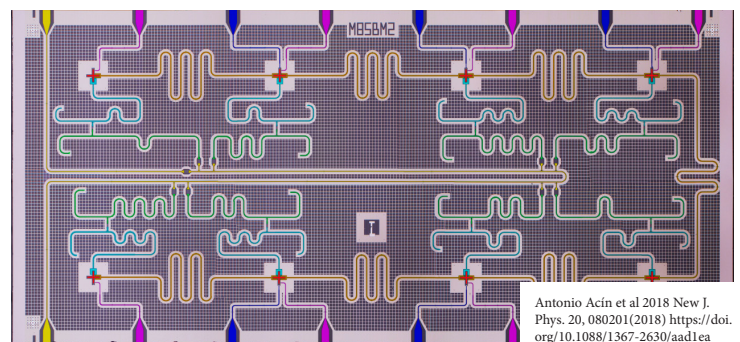
Global quantum network



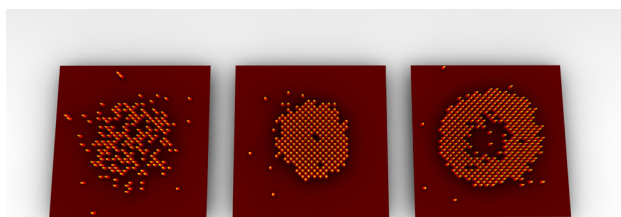
Entangled photon generation



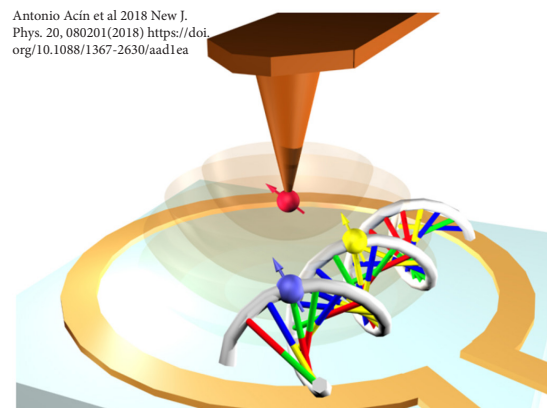
Single nanoemitter



Superconducting quantum processor



Quantum microscopy



Quantum biosensing



THE CONFERENCE MAIN 4 CATEGORIES AND SUBCATEGORIES

The Conference will also host a round table, open to all participants on:

QUANTUM MECHANICS AND BEYOND

1. QUANTUM MATERIALS

Topological Materials
(e.g., topological insulators and Weyl semi-metals)
2D Van der Waals materials
(e.g. Graphene, TMDCs, 2D magnets)
Carbon-based Materials
Antiferromagnetic Materials (e.g., Non-collinear
Ferromagnets, Ferrimagnets, Skyrmions)

Nano-Photonics, Metamaterials, Active Plasmonics
Nano-Structured and Functionalized Surfaces
Phononic Materials
Spin-orbitronics
Multiferroics
Superconducting Oxides
Semiconductor Heterostructure

2. QUANTUM DEVICES

Semiconductor Quantum Cascade lasers
Quantum Dots Devices
Wide-bandgap Solar-blind Photodetectors & Focal
Plane Arrays
Terahertz Emitters and Receivers
Terahertz Nano-Photonics and Nanoelectronics
Single-photon/ Pair-photons Counting Detectors
Biosensing - Bio-Electronics - Bio-Photonics
Spintronic Gigahertz to Terahertz Sources and
Detectors

Magnetic Memory and Logic Devices
Topological Spintronic Devices
Quantum Memories
Magnonic Devices
Valleytronic Devices
Nano-antenna Devices
Nano-optomechanics
Quantum Cascade Laser Frequency Combs

3. CHARACTERIZATION, ANALYSIS AND SPECTROSCOPY OF QUANTUM MATERIALS AND DEVICES

Quantum Spectroscopy and Sensing
Nonlinear and Ultrafast Optical Phenomena
Materials characterization
(Optical, Electrical, Magnetic, Structural, etc.)
Near-Field Optics and Flat Optics
Light-Matter Interaction at the Nano-Scale

Quantum Optical Measurements and Decoherence
Effects
Novel Magnetic Microscopy Techniques
Magneto-Optics
Quantum Metrology

4. SYSTEMS AND APPLICATIONS

Unconventional Quantum Computing
Quantum Communications and Applications
Integrated Photonics for Quantum Sensing and
Quantum imaging
Neuromorphic Devices, Circuits, and Systems

Micro- and Nano-Fabrication Technologies
Heterogenous Integration Technologies
Photonic Bandgap Structures and their Applications
Magnonics Applications



Local Scientific Committee Members:

- Frederic Grillot, Telecom Paris, France
- Heming Huang, Telecom Paris, France
- Juan Alvarez, Telecom Paris, France
- Mingzhao Shi, Telecom Paris, France
- Nicolas Fabre, Telecom Paris, France
- Sara Zaminga, Telecom Paris, France

International Scientific Committee Members:

- Arnaud Brignon, Thales Research and Technology, Paris, France
- Baptiste Fix, ONERA Paris, France
- Carlo Sirtori, ENS Paris, Paris, France
- Daniel Dolfi, Thales Research and Technology, Paris, France
- Martin Defour, Thales Research and Technology, Paris, France
- Feihu WANG, Quantum Science Center of Guangdong-Hong-Kong-Macao Greater Bay Area, China
- Ferechteh H. Teherani, Nanovation, France
- Jean-Pascal Duchemin, Thales Research and Technology, France
- Jerzy M. Langer, Polish Academy of Sciences, Poland
- Paul-Louis Meunier, ESTP-Paris, France
- Minh Nguyen, HRL Laboratories, LLC, USA
- Olivier Acher, HORIBA France SAS, Palaiseau, France
- Quanyong Lu, Beijing Academy of Quantum Information Sciences, China
- Selim M. Shahriar, Northwestern University, USA
- Steve Slivken, Northwestern University, USA
- Jean-Pierre Huignard, Institut Polytechnique de Paris, France
- Fariba HATAMI, Humboldt-Universität Zu Berlin, Germany
- Detlev Gruetzmacher, Forschungszentrum Jülich GHMB, Germany



Keynote Speakers



Title: Electromagnetic field sensing with Rydberg atoms

Sylvain SCHWARTZ

Université Paris-Saclay, France

Title: A unified model for light emission from solids

Jean-Jacques Greffet

Université Paris-Saclay, CNRS, Palaiseau, France



Title: Will be updated soon

Agnès Maître

Sorbonne University, France



Hot Topics and Latest News presented by Experts



Title: Antimonide-based Narrow Bandgap Semiconductors for Infrared Technology and Quantum Information Science
Binh-Minh Nguyen
HRL Laboratories, LLC, Malibu, California, USA

Title: Silicon-Rich Nitride Nonlinear Nanophononics
Yeshiahu Fainman
University of California San Diego, USA



Title: Topological Superconductivity and Associated Majorana States: Towards Fault-Tolerant Qubits for Topological Quantum Computation
Roland Wiesendanger
University of Hamburg, Germany

Title: Detection-Loophole-Free Bell Tests over Long Distances
Yazeed K. Alwehaibi
Imperial College London, London, United Kingdom



Title: Quantum Optical Nature of High-Harmonic Generation in Semiconductors
Mr. David Theidel
École nationale supérieure de techniques avancées (ENSTA) ParisTech, CNRS, France

Title: Quantum computational machine vision
Alexander Lvovsky
University of Oxford, United Kingdom



Title: Sub 5 picosecond resolution time-correlated single photon counting system with demountable radio frequency photomultiplier tube
Amur Margaryan
A.I. Alikhanyan National Science Laboratory (Yerevan Physics Institute), Yerevan, Armenia

Title: Many-Body Plasmonic Platforms for Quantum Sensing
Omar Magana-Loaiza
Louisiana State University, USA



Title: Topological Insulator in InN/InGaN Quantum Wells
Sławomir Paweł Łepkowski
Polish Academy of Sciences, Poland

Title: Boron nitride in the run-up to quantum devices
Lorenzo Sponza
Université Paris-Saclay, France



Title: Nanophysics as an Innovation Discovery
Vladimir Valentinovich Egorov
National Research Center "Kurchatov Institute, Russia

Title: Nano-/micro-prism-based IV-VI Lead Selenide photoconductive detector
Johannes Kunsch
Laser Components Germany GmbH, Germany



Title: 2D twisted BN: a tunable two-level quantum system
Hakim Amara
Université Paris-Saclay, ONERA, CNRS, France

Title: Vacuum-one-photon encoded states with semiconductor quantum dots
Carlos Anton Solanas
Universidad Autónoma de Madrid, Spain



Title: Quantum cooperation in multiphoton conversion during laser scattering in multiple cascades and its application in information processing
Nicolae A. Enaki
Institute of Applied Physics of Moldova State University, Republic of Moldova

Title: Mid-Infrared Ring Laser Platforms for Compact Broadband Gas Sensing
Rolf Szedlak
Institute of Solid State Electronics, Austria



Title: Improving the bandwidth of Rydberg RF receivers: theory and experiments
Fabien Bretenaker
Université Paris-Saclay, CNRS, France

Title: Chemically Synthesized Core/Shell and Core/Shell/Shell Infrared Nanocrystals for Quantum Devices
Tischler Joseph G
The University of Oklahoma, USA



Title: High power THz sources based on quantum cascade lasers
Quanyong Lu
Beijing Academy of Quantum Information Sciences, China

Title: Mid Infrared Silicon Photonics
Mauro Fernandes Pereira
Khalifa University, Abu Dhabi, UAE





Hot Topics and Latest News presented by Experts



Title: Thales on the way toward Quantum Computing applications at scale based on Applications-oriented Benchmark for Quantum Computer
Frédéric BARBARESCO
THALES Land & Air Systems, Velizy-Villacoublay, France

Title: Advances in AlGaAs photonics chips for quantum metrology and quantum communications
Sara Ducci
Université Paris Cité, CNRS, France



Title: Nanostructure Engineered III-V Quantum Dots for Telecom and Mid-Infrared Single Photon Sources
John Prineas
University of Iowa, USA

Title: Bosonic Quantum Information from Superselection Rules
Pérola Milman
CNRS, Université Paris Cité, France



Title: Recent progress on antenna-coupled, mid-infrared unipolar photodetectors: high-speed operation and two-photon detection
Stefano Barbieri
Laboratoire IEMN and CNRS, France

Title: Near field imaging and spectroscopy of complex stacking of semiconductors
Baptiste FIX
DOTA, ONERA, Université Paris Saclay, France



Title: Photonic Integrated Visible to SWIR Stabilized Lasers and Control for Neutral Atom and Trapped Ion Quantum Timing and Sensing
Dan Blumenthal
University of California Santa Barbara, USA

Title: Cold atom inertial quantum sensors: addressing dead time issues
Arnaud Landragin
Université PSL, Sorbonne Université, CNRS, France



Title: Electron-nuclear entanglement dynamics and photonic resource state generation from quantum dots
Edwin Barnes
Virginia Tech, Blacksburg, VA USA

Title: Signal state engineering in continuous-variable quantum communication
Vladyslav C. Usenko
Palacky University, Czech Republic.



Title: Integration of rigorous optimization and machine learning tools for designing large-scale multi-functional nanophotonic structures
Adibi Ali
Georgia Institute of Technology, USA

Title: Quantum enhanced detection for gravitational waves using atomic spin ensembles
Maimouna Bocoum
CNRS-ESPCI, France



Title: Will be updated soon
Paolo Villoresi
University of Padova, Italy

Title: Will be updated soon
Merdji Hamed (M.)
École Polytechnique, France



Title: Will be updated soon
Daniele Bajoni
Photonics Group Unipv, Italy

Title: Will be updated soon
Malik Mehul
Heriot-Watt University, UK



Title: Standards and Certification in Quantum Communications: Recent Advances and Challenges
Hao Qin
National University of Singapore, Singapore



Hot Topics and Latest News presented by Experts



Title: Will be updated soon
Hiroshi Ito
Kyushu Institute of Technology, Japan

Title: Will be updated soon
Nadia Belabas
Université Paris-Saclay, France



Title: Resources and applications for advanced quantum networking
Eleni Diamanti
Sorbonne Université, France

Title: Will be updated soon
Annick Loiseau
ONERA, CNRS, France



Title: Will be updated soon
Fariba Hatami
Humboldt University of Berlin, Germany

Title: Will be updated soon
Economou, Sophia
Virginia Tech university, USA



Title: Will be updated soon
Kamel Bencheikh
Université Paris-Saclay, France

Title: Will be updated soon
Julien Laurat
Sorbonne Université, France



Title: Will be updated soon
Dafiné Ravelosona

Title: Will be updated soon
Romain Alléaume
Telecom Paris, France



Title: Will be updated soon
Sébastien Tanzilli
CNRS & Université Côte d'Azur, France

Title: Will be updated soon
Debuisschert Thierry
Thales Group, France



Title: Will be updated soon
Stephane Guerin
Université Bourgogne, France

Title: Will be updated soon
Tobias Kippenberg
EPFL, Switzerland



Title: Will be updated soon
Michael Kues
Leibniz University Hannover, Germany

Title: Will be updated soon
Andrea Ivano Melloni
Politecnico di Milano, Italy



Title: Will be updated soon
Konrad Banaszek
University of Warsaw, Poland

Title: Will be updated soon
Siddarth Joshi
University of Bristol, UK



Title: Will be updated soon
Michał Karpiński
University of Warsaw, Poland



Hot Topics and Latest News presented by Experts



Title: Will be updated soon
Carlos Ramos
Université Paris-Saclay, France

Title: Will be updated soon
Sébastien Bidault
Institut Langevin, ESPCI ParisTech, France



Title: Will be updated soon
Dan Buca
Research Center Jülich GmbH, Germany

Title: Will be updated soon
Selim M Shahriar
Northwestern University, USA



Title: Will be updated soon
N. S. Agarwal
India

Title: Will be updated soon
S. Ginil Mon
Nesamony Memorial Christian College, India



Title: Will be updated soon
A.K. Mishra
India

Title: Will be updated soon
Siddhartha Bhattacharyya
VSB Technical University of Ostrava, Czech Republic



Title: Will be updated soon
Carmine Antonio Perroni
Università "Federico II" Napoli, Italy

Title: Will be updated soon
Daniele Parlato
Thales Group, France



Title: Will be updated soon
A.S. Sitdikov
Russia

Title: Will be updated soon
Tran Thi Kim Ngan
Thailand



Title: Will be updated soon
Florin Lucian CONSTANTIN
Université de Lille, France

Title: Will be updated soon
Vladyslav Usenko
Palacky University Olomouc, France



Title: Will be updated soon
Alice Sinatra
Sorbonne Université, France

Title: Will be updated soon
Lorenzo Columbo
Université Paris-Saclay, ONERA, Paris, France





Posters



Title: Selective band engineering of Bi/Si(111) by boron segregation

Lorenzo Sponza

Université Paris-Saclay, France



Title: Toward Spin Liquids: Monte Carlo Study of The XY Model on a Kagome Bilayer

Norah Abdulaziz A Alsayegh

Queen Mary University of London, UK

More details will be updated soon

For Details Visit: <https://quest-conference.com/sponsors.php>

For Abstract Submission

<https://www.quest-conference.com/abstract.php>

For Registration

<https://www.quest-conference.com/registration.php>

Conference Secretary:

questconference86@gmail.com | quantum@quest-conference.com

<https://www.quest-conference.com/>